

Cross docking med fokus på door assignment



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Introduction

Flow of goods

Movements

Doors and
distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

Introduction

- ▶ A cross dock serves as a terminal for consolidating freights
- ▶ It has the role of a transshipment point in transportation
- ▶ Distribution centres function as cross docks



Introduction

Flow of goods

Movements

Doors and
distances

Trucks and doors

CDAP elements

QAP

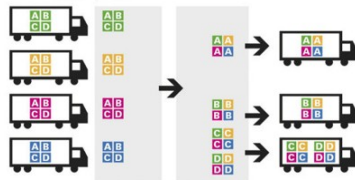
CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

Flow of goods

- ▶ Goods arrive on inbound trucks and leave on outbound trucks
- ▶ Each inbound truck delivers goods from a single given origin
- ▶ Each outbound truck delivers goods to one or more given destinations



Introduction

Flow of goods

Movements

Doors and
distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

Movements inside the cross dock

- ▶ Forklifts move pallets between doors
- ▶ Only one pallet is moved at a time by any forklift
- ▶ The workload for moving a pallet can be described as the distance that it must be moved



Introduction

Flow of goods

Movements

Doors and
distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

Doors and distances



In this example, the distance between doors is described as the vertical distance in the picture between the two doors, resulting in the following distance matrix:

Distances between doors:

	01	02	03	04	05
I1	2	2	6	10	14
I2	6	2	2	6	10
I3	10	6	2	2	6
I4	14	10	6	2	2

Introduction

Flow of goods

Movements

Doors and
distances

Trucks and doors

CDAP elements

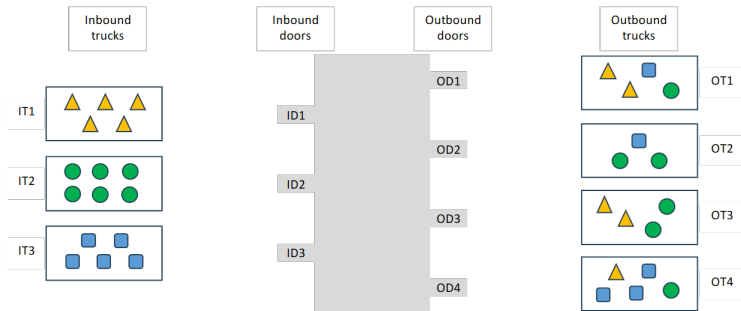
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CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

Trucks and doors



Using the crossdock facilities, the goods are moved from inbound trucks via inbound doors to outbound doors and placed on outbound trucks according to the demanded quantities at the destinations

Introduction

Flow of goods

Movements

Doors and distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

The Cross-dock Door Assignment Problem (CDAP): Parameters

- ▶ M is the set of inbound trucks (origin nodes)
- ▶ N is the set of outbound trucks (destination nodes)
- ▶ I is the set of inbound doors
- ▶ J is the set of outbound doors
- ▶ S_i is the capacity of inbound door i
- ▶ R_j is the capacity of outbound door j
- ▶ d_{ij} is the distance from inbound door i to outbound door j
- ▶ w_{mn} is the volume (flow) from inbound truck m to outbound truck n
- ▶ $s_m = \sum_{n \in N} w_{mn}$ is the total volume from inbound truck m
- ▶ $r_n = \sum_{m \in M} w_{mn}$ is the total volume to outbound truck n

Introduction

Flow of goods

Movements

Doors and
distances

Trucks and doors

CDAP elements

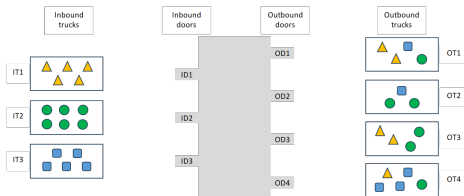
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CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

CDAP: Decisions and decision variables



Decisions: Assign inbound trucks to inbound doors and outbound trucks to outbound doors

$$x_{mi} = \begin{cases} 1 & \text{if inbound truck } m \text{ is assigned to inbound door } i \\ 0 & \text{otherwise} \end{cases}$$

$$y_{nj} = \begin{cases} 1 & \text{if outbound truck } n \text{ is assigned to outbound door } j \\ 0 & \text{otherwise} \end{cases}$$

Objective: Minimize total material handling cost (specified on the next slide)

Introduction

Flow of goods

Movements

Doors and distances

Trucks and doors

CDAP elements

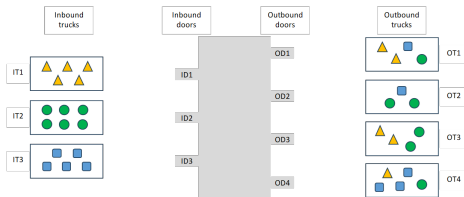
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CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

Material handling cost



For each pair of inbound truck m and outbound truck n , the associated material handling cost is obtained as the product of volume and distance:

- ▶ The volume that must be moved from truck m to truck n . This value is given as part of the input
- ▶ For example, in the picture above, there is a volume of $w_{13} = 2$ units (2 triangles) from IT1 to OT3
- ▶ If IT1 is assigned to ID2 and OT3 is assigned to OD4, each of the 2 units must be moved $d_{24} = 6$ distance units
- ▶ As such, with these two assignments, there is a cost of $2 \times 6 = 12$ from IT1 to OT3

The total material handling cost is obtained as the material handling cost summed over all pairs of inbound and outbound trucks

Introduction

Flow of goods

Movements

Doors and
distances

Trucks and doors

CDAP elements

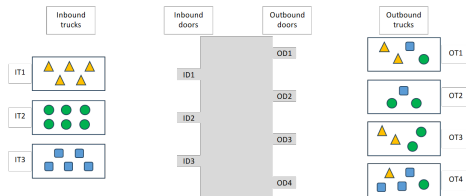
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CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

CDAP: A special case



Suppose the following holds:

- ▶ Each IT must be assigned to one ID, and each ID must be assigned to one IT
- ▶ Each OT must be assigned to one OD, and each OD must be assigned to one OT

Then we have an assignment problem on each side of the cross dock, and the CDAP can be formulated as a Quadratic Assignment Problem (QAP)

Introduction

Flow of goods

Movements

Doors and distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

CDAP: Formulation of the special case as a QAP

$$\min. \quad \sum_{m \in M} \sum_{i \in I} \sum_{n \in N} \sum_{j \in J} d_{ij} w_{mn} x_{mi} y_{nj}$$

$$\text{s.t.} \quad \sum_{m \in M} x_{mi} = 1, \forall i \in I \quad (\text{One inbound truck to inbound door } i)$$

$$\sum_{i \in I} x_{mi} = 1, \forall m \in M \quad (\text{One inbound door to inbound truck } m)$$

$$x_{mi} \in \{0, 1\}, \forall m \in M, i \in I \quad (\text{Binary } x \text{ variables})$$

$$\sum_{n \in N} y_{nj} = 1, \forall j \in J \quad (\text{One outbound truck to outbound door } j)$$

$$\sum_{j \in J} y_{nj} = 1, \forall n \in N \quad (\text{One outbound door to outbound truck } n)$$

$$y_{nj} \in \{0, 1\}, \forall n \in N, j \in J \quad (\text{Binary } y \text{ variables})$$

Introduction

Flow of goods

Movements

Doors and
distances

Trucks and doors

CDAP elements

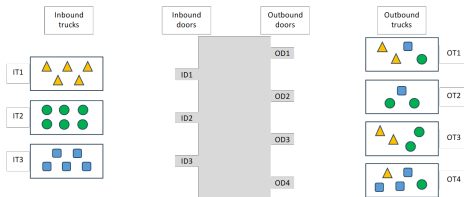
QAP

CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

CDAP: Considering door capacities



- ▶ Instead of the QAP special case, the problem becomes (even) more complicated when door capacities are considered
- ▶ Inbound trucks must be assigned to inbound doors subject to capacities of inbound doors
- ▶ Outbound trucks must be assigned to outbound doors subject to capacities of outbound doors

We then obtain a formulation of CDAP which is more general than the QAP, as shown on the following slide

Introduction

Flow of goods

Movements

Doors and distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

CDAP: Formulation

$$\min. \quad \sum_{m \in M} \sum_{i \in I} \sum_{n \in N} \sum_{j \in J} d_{ij} w_{mn} x_{mi} y_{nj}$$

$$\text{s.t.} \quad \sum_{m \in M} s_m x_{mi} \leq S_i, \forall i \in I \quad (1a) \text{ (Capacity constraint for inbound door } i)$$

$$\sum_{i \in I} x_{mi} = 1, \forall m \in M \quad (1b) \text{ (One inbound door to inbound truck } m)$$

$$x_{mi} \in \{0, 1\}, \forall m \in M, i \in I \quad (1c) \text{ (Binary } x \text{ variables)}$$

$$\sum_{n \in N} r_n y_{nj} \leq R_j, \forall j \in J \quad (2a) \text{ (Capacity constraint for outbound door } j)$$

$$\sum_{j \in J} y_{nj} = 1, \forall n \in N \quad (2b) \text{ (One outbound door to outbound truck } n)$$

$$y_{nj} \in \{0, 1\}, \forall n \in N, j \in J \quad (2c) \text{ (Binary } y \text{ variables)}$$

- ▶ In the following, $\text{GAP}(x, S)$ refers to the constraint set (1a), (1b) and (1c), and $\text{GAP}(y, R)$ refers to the constraint set (2a), (2b) and (2c)
- ▶ $\text{GAP}(x, S)$ are the constraints of a Generalized Assignment Problem (GAP) on the inbound side
- ▶ $\text{GAP}(y, R)$ are the constraints of a Generalized Assignment Problem (GAP) on the outbound side
- ▶ This leads to a simplified formulation as shown on the next page

Introduction

Flow of goods

Movements

Doors and distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

CDAP: Simplified formulation

Given the introduced notation, the CDAP can be formulated as follows:

$$\begin{array}{ll}\min. & \sum_{m \in M} \sum_{i \in I} \sum_{n \in N} \sum_{j \in J} d_{ij} w_{mn} x_{mi} y_{nj} \\ \text{s.t.} & \text{GAP}(x, S) \quad \quad \quad (\text{inbound assignments}) \\ & \text{GAP}(y, R) \quad \quad \quad (\text{outbound assignments})\end{array}$$

Introduction

Flow of goods

Movements

Doors and
distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

Local Search Heuristic

- ▶ We consider again the CDAP and the following formulation:

$$\begin{aligned} \min. \quad & \sum_{m \in M} \sum_{i \in I} \sum_{n \in N} \sum_{j \in J} d_{ij} w_{mn} x_{mi} y_{nj} \\ \text{s.t.} \quad & \text{GAP}(x, S) && (\text{inbound assignments}) \\ & \text{GAP}(y, R) && (\text{outbound assignments}) \end{aligned}$$

- ▶ For given values $\hat{y}$ of y variables satisfying $\text{GAP}(y, R)$, the remaining problem is the following:

$$\begin{aligned} \min. \quad & \sum_{m \in M} \sum_{i \in I} \left[\sum_{n \in N} \sum_{j \in J} d_{ij} w_{mn} \hat{y}_{nj} \right] x_{mi} \\ \text{s.t.} \quad & \text{GAP}(x, S) \end{aligned}$$

This is a Generalised Assignment Problem with only x variables as decision variables

- ▶ Similarly, for given values $\hat{x}$ of x variables satisfying $\text{GAP}(x, S)$, the remaining problem is the following:

$$\begin{aligned} \min. \quad & \sum_{n \in N} \sum_{j \in J} \left[\sum_{m \in M} \sum_{i \in I} d_{ij} w_{mn} \hat{x}_{mi} \right] y_{nj} \\ \text{s.t.} \quad & \text{GAP}(y, R) \end{aligned}$$

This is a Generalised Assignment Problem with only y variables as decision variables

Introduction

Flow of goods

Movements

Doors and distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search Heuristic

Greedy Construction Heuristic

Local Search Heuristic, continued

Local Search Heuristic for the CDAP

```
1  Generate an initial solution  $\hat{y}$  to the outbound assignments
2  Set  $B = \infty$  // Best known solution
3
4  repeat
5      Optimize  $x$  for the given  $\hat{y}$ , resulting in  $\hat{x}$ 
6      Optimize  $y$  for the given  $\hat{x}$ , resulting in  $\hat{y}$  and objective value  $z$ 
7      if  $z < B$  then
8          |    $z = B$  // Improvement: Update best known solution
9      end
10 until no improvement
```

- ▶ The LSH repeatedly optimizes one side of the assignments at a time, alternating between inside and outside assignments, for a given assignment on the other side
- ▶ The initial $\hat{y}$ solution can be obtained by choosing randomly a door for each truck. As such, this initial $\hat{y}$ does not have to satisfy the capacity constraints on the outbound doors
- ▶ The entire LSH can be done repeatedly by starting from different random solutions in line 1 and improving each initial solution. The final solution is then the best of all improved solutions
- ▶ Alternatively, the LSH can be started by an initial solution to the inbound assignments in line 1
- ▶ Note that each GAP optimization problem in lines 5–6 can be difficult to solve. As such, it can be meaningful to consider heuristics for these partial problems as well

Introduction

Flow of goods

Movements

Doors and
distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic

A greedy construction heuristic for inbound assignments

- ▶ In the Local Search Heuristic, only one side of the assignments is considered at a time. Given that the problem on only one side is a Generalized Assignment Problem (GAP), we can consider heuristic approaches to the GAP
- ▶ In the separate material on packing problems, a greedy construction heuristic by Martello & Toth was mentioned. In the following, this heuristic is adapted to the inbound GAP, which was described as follows on a previous slide:

"For given values $\hat{y}$ of y variables satisfying $\text{GAP}(y, R)$, the remaining problem is the following:

$$\begin{aligned} \min. \quad & \sum_{m \in M} \sum_{i \in I} \left[\sum_{n \in N} \sum_{j \in J} d_{ij} w_{mn} \hat{y}_{nj} \right] x_{mi} \\ \text{s.t.} \quad & \text{GAP}(x, S) \end{aligned}$$

This is a Generalised Assignment Problem with only x variables as decision variables"

- ▶ To simplify notation, we can introduce c_{mi} as the cost of assigning truck m to door i , where c_{mi} can be calculated as follows:

$$c_{mi} = \sum_{n \in N} \sum_{j \in J} d_{ij} w_{mn} \hat{y}_{nj}$$

Introduction

Flow of goods

Movements

Doors and distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search Heuristic

Greedy Construction Heuristic

A greedy construction heuristic, continued (I)

- Given the introduced notation on the previous slides, the inbound GAP can be described as follows:

$$\min. \quad \sum_{m \in M} \sum_{i \in I} c_{mi} x_{mi}$$

$$\text{s.t.} \quad \sum_{m \in M} s_m x_{mi} \leq S_i, \forall i \in I \quad (1a) \text{ (Capacity constraint for inbound door } i)$$

$$\sum_{i \in I} x_{mi} = 1, \forall m \in M \quad (1b) \text{ (One inbound door to inbound truck } m)$$

$$x_{mi} \in \{0, 1\}, \forall m \in M, i \in I \quad (1c) \text{ (Binary } x \text{ variables)}$$

Introduction

Flow of goods

Movements

Doors and distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search Heuristic

Greedy Construction Heuristic

A greedy construction heuristic, continued (II)

Greedy construction for inbound GAP

```
// This is based on the GAP heuristic by Martello & Toth
// Initially, all trucks are unassigned
```

```
1 repeat
2   if there is an unassigned truck with no feasible doors then
3     // Stop. No feasible solution is found
4   else if there is an unassigned truck  $m$  with only one feasible door  $i$  then
5     Assign truck  $m$  to door  $i$ 
6   else
7     For each unassigned truck  $m$ , compute  $\delta_m = c''_m - c'_m$ , where  $c'_m$  is the smallest and  $c''_m$ 
8     is the second smallest assignment cost  $c_{mi}$  among all feasible doors for truck  $m$ 
9     Choose the unassigned truck  $m$  with the largest  $\delta_m$  and assign it to the door  $i$  with
      smallest cost  $c'_m$ 
8   end
9 until all trucks are assigned
```

- ▶ In each iteration, after assigning truck m to door i , the following updates are made:
 - ▶ The remaining capacity for door i is reduced by s_m
 - ▶ Truck m is removed from the set of unassigned trucks

Introduction

Flow of goods

Movements

Doors and
distances

Trucks and doors

CDAP elements

QAP

CDAP

Local Search
Heuristic

Greedy
Construction
Heuristic